

FALAN LATIN AMERICAN TRAINING PROGRAM 2026

WEEK 1

- Monday - Wednesday and Friday, Auditorio Facultad de Química y Biología, Universidad de Santiago de Chile

- Thursday, Sala BNI, Facultad de Medicina Norte, Universidad de Chile

	Monday August 10	Tuesday August 11	Wednesday August 12	Thursday August 13	Friday August 14
9:00-10:00		Ion channels and membrane excitability Dr. Patricio Rojas (Universidad de Santiago de Chile)	Optogenetic investigation of neural circuits <i>in vivo</i> Dr. Elias Leiva (Universidad de Santiago de Chile)	EEG for Neuroscience Research (I) Dr. Boris Lucero (Universidad Católica del Maule)	
10:00-11:00	Opening Session IBRO/ISN Presentation Host Organizing Committee members	The patch-clamp technique Dr. Fernando Ortiz (Universidad de Santiago de Chile)	<i>In vivo</i> electrophysiological recordings Dr. José Luis Valdés (Universidad de Chile)	EEG for Neuroscience Research (II) Dr. Christ Devia (Universidad de Chile)	Students' presentations
11:00-11:30	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
11:30-12:30	Ion channels as the doors of perception Dr. Ramón Latorre Dr. Francisco Bezanilla (CINV)	Ion channels in thermotransduction Dr. Rodolfo Madrid (Universidad de Santiago de Chile)	Integrative Neuroscience Approaches Dr. Alexia Nunez-Parra (Universidad de Chile)	Active perception in the visual system, Dr. Pedro Maldonado (Universidad de Chile)	Students' presentations
12:30-14:00	Lunch Break	Lunch Break	Lunch Break	Lunch Break	Lunch Break
14:00-	<i>Hands-on</i>	<i>Hands-on</i>	<i>Hands-on</i>	<i>Hands-on</i>	<i>Hands-on</i>

18:00	Workshop	Workshop	Workshop	Workshop	Workshop
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WEEK 2

Activities will be at Auditorio Facultad de Ciencias, Universidad de Valparaíso

	Monday August 17	Tuesday August 18	Wednesday August 19	Thursday August 20	Friday August 21
9:00-10:00	Principles of Neural Modeling: from Neurons to Networks I Dr. Patricio Orio (Universidad de Valparaíso)	Whole-brain modeling and applications I Dr. Wael El-Deredy (Universidad de Valparaíso)	Structural Brain Connectivity Analysis Dr. Pamela Guevara (Universidad de Valparaíso)	The Vulnerable Adolescent Brain: Parvalbumin Interneuron Dysfunction in Neuropsychiatric Disorders Dr. Marco Fuenzalida (CENFI, Universidad de Valparaíso) The fruit fly <i>Drosophila melanogaster</i> as a model organism for studying Autism Spectrum Disorder" Dr. Angelina Palacios (CINV, CICOM, Universidad de Valparaíso)	Neuromodulation at central synapses in health and disease Dr. Andrés Chávez (CINV, Universidad de Valparaíso) Functional Crosstalk Between $\alpha 7$ nAChRs and P2X7 Receptors: Implications for microglia activation Dra. Constanza Maldifassi (CINV, Universidad de Valparaíso)
10:00-11:00	Principles of Neural Modeling: from Neurons to Networks II Dr. Patricio Orio (Universidad de Valparaíso)	Whole-brain modeling and applications II Dr. Mónica Otero (Universidad San Sebastián)	Entropy and complexity analyses of electrophysiological signals: Application to Alzheimer's disease detection in the eye" I	Impact of dopamine levels and D2 receptors on the generation of compulsive behaviors, Dra. Angélica Escobar	Pannexin 1 in Alzheimer disease, bridging A beta toxicity, neuro inflammation and synaptic dysfunction Dr. Álvaro Ardiles

	Valparaíso)		Dr. Leonel Medina (Universidad de Santiago de Chile)	(CENFI, Universidad de Valparaíso) Glutamatergic alterations underlying compulsive behavior” Dr. Pablo Moya (CINV, C-ESTRES, Universidad de Valparaíso)	(CINV, CiTne, Universidad de Valparaíso) The Olfactory Sensory System as a Window into Inflammation Based Dementia Dra. Kathleen Whitlock, (CINV, Universidad de Valparaíso)
11:00-11:30	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
11:30-12:30	Exploring Basal Ganglia Function Through Computational Modeling Dr. Javier Baladron (Universidad de Santiago de Chile)	Whole Brain modelling and Brain oscillations Dr. Felipe Torres (Universidad Católica del Maule)	Entropy and complexity analyses of electrophysiological signals: Application to Alzheimer's disease detection in the eye” II. Dr. Leonel Medina. (Universidad de Santiago de Chile)	Neurobiology of stress resilience and depression: Rol of GR/MR balance, Dr. Alexies Dagnino (CIES, Universidad de Valparaíso) Effects of chronic exposure to a high-fat diet on the lateral septum-lateral hypothalamus pathway Dr. Ramón Sotomayor (CENFI, Universidad de Valparaíso)	A bone drug with a twist: Zoledronic acid eases pain through TRPV1 inhibition, Dra. Karen Castillo (CINV, Universidad de Valparaíso) Role of hemichannels in caries-mediated dentinal hypersensitivity Dr. Isaac García (CINV, CICOM, Universidad de

					Valparaíso)
14:00-17:00	<i>Hands-on Workshop</i>	<i>Hands-on Workshop</i>	<i>Hands-on Workshop</i>	<i>Hands-on Workshop</i>	
17:00-18					Keynote Lecture. Properties of large pore channels and their role in animal models of nervous system diseases. Dr. Juan Carlos Saez (CINV, Universidad de Valparaíso)

WEEK 3

- Monday, Tuesday and Friday, Pontificia Universidad Catolica de Chile

- Wednesday, Thursday, Sala BNI, Facultad de Medicina, Universidad de Chile

- Friday, Universidad de Santiago de Chile

	Monday August 24	Tuesday August 25	Wednesday August 26	Thursday August 27	Friday August 28
9:00-10:00	Molecular Mechanism of synaptic dysfunction in neurodegeneration Dr. Alejandra Álvarez. (Pontificia Universidad Católica de Chile)	Molecular and cellular mechanisms mediating sensory transmission Dr. Evelyn Avilés Dr. Margarita Calvo (Pontificia Universidad Católica de Chile)	Bulk short-read and long-read sequencing for isoform discovery Dr. Christine H. Tabuloc (University of California, Davis)	To be confirmed. Multi-omics approaches in biomedicine Dr. Lucia Peixoto (Washington State University)	Adaptive optics for <i>in vivo</i> brain Dr. Cristina Rodríguez (Yale University)
10:00-11:00	Intracellular trafficking of membrane proteins in neurons Dr. María Paz Marzolo (Pontificia Universidad Católica de Chile)	Molecular bases of the nervous system development Dr. Ignacio Cancino Dr. Claudio Perez- Leighton (Pontificia Universidad Católica de Chile)	Single-cell sequencing Dr. Vinicius Maracaja-Coutinho (Universidad de Chile)	Spatial transcriptomics in neuroscience Dr. Xiangmin Xu (University of California, Irvine)	Optogenetic manipulation of visual perception Dr. Luis Carrillo (Universidad Nacional Autónoma de México)
11:00-11:30	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
11:30-12:30	Intravital microscopy techniques for monitoring brain activity and metabolism Dr. Víctor Hugo Cornejo Dr. Andrés González (Pontificia Universidad Católica de Chile)	Molecular and cellular mechanisms mediating sensory transmission. Dra. Margarita Calvo. (Pontificia Universidad Católica de Chile)	Transcriptional profiling of cell lineages of sex-differentiated neurons Dr. E. Josie Clowney (University of Michigan)	Integration of spatial and single-cell transcriptomics datasets Dr. Sergio Hidalgo (Washington State University)	Title to be confirmed Dr. Xiangmin Xu (University of California, Irvine)
14:00-18:00	<i>Hands-on Workshop</i>	<i>Hands-on Workshop</i>	<i>Hands-on Workshop</i>	<i>Hands-on Workshop</i>	<i>Professional Development Workshop.</i>

Hands-on Workshops 1st week

- Synaptic Plasticity in the Hippocampus (**Carlos Rozas**, Universidad de Santiago de Chile)
- Brain activity using fluorescence (**Alexia Nuñez, Jorge Vera**, Facultad de Ciencias, Universidad de Chile)
- *In vitro* Ca²⁺ -Imaging (**Maria Pertusa, Rodolfo Madrid**, Universidad de Santiago de Chile)
- Measurement of the C Reflex (**Luis Constandil**, Universidad de Santiago de Chile)
- Physics of Electrical Neurostimulation (**Leonel Medina, Cristian Morales, Rodrigo Osorio**, Universidad de Santiago de Chile)
- How to Set Up a “Dirt-Cheap” Electrophysiology Rig (**Patricio Rojas, Cristian Moreno**, Universidad de Santiago de Chile)
- EEG Analysis During Continuous Speech Perception Using Multivariate Temporal Response Functions (mTRF) (**Ana Campos**, Universidad de Chile, Medicina Norte)
- Two-electrode voltage-clamp of the neuromuscular *Drosophila* synapse (**Ricardo Delgado**, Facultad de Ciencias, Universidad de Chile)

Hands-on Workshops 2nd week: (Universidad de Valparaíso)

- Behavioral assays in the fruit fly *Drosophila melanogaster* (**Angelina Palacios-Muñoz**)
- Electrophysiology: Interneurons and GABA Transmission (**Marco Fuenzalida Núñez**)
- Basic simulation of neural excitability (**Patricio Orio**)
- Stress Resilience (**Alexies Dagnino**)
- Introduction to the use and analysis of Multielectrodes Array (**Adrian Palacios**)
- Chemogenetic inhibition of Amígdala-NAcc projection (**Dra. Angélica Escobar / Dr. Jorge Mallea**)
- Visualization of Neurons in Live Animals (**Kathleen Whitlock**)
- Use of inducible genetic systems to study animal behavior (**John Ewer**)
- Whole-brain modelling (**Mónica Otero / Fernando Lehue**)
- Neural activity from freely behaving mice (**Ignacio Negrón-Oyarzo**)
- Techniques for Measuring Brain Neurotransmitters (**Ramón Sotomayor-Zárate**)
- Neurokit2 tutorial: Hands-on entropy and complexity estimation (**Leonel Medina**)

Hands-on Workshops 3rd week:

- *In vivo* Microdialysis in Rat or Mouse Models, Including Stereotaxic Surgery and Live Rat Demonstration (**Maria Estela Andres, Claudio Perez-Leighton**, Pontificia Universidad Católica de Chile)
- Preparation of Hippocampal Neuron Cultures and Analysis of Dendritogenesis (**Alejandra Alvarez**, Pontificia Universidad Católica de Chile)
- Studying Central Nervous System Development Using Brain Organoids (**Ignacio Cancino, Francisca Cornejo**, Pontificia Universidad Católica de Chile)
- Learning About the Nervous System and Locomotor Behavior in Invertebrate Models (*C. elegans* and *Drosophila*) (**Loreto Véliz, Alicia Nogueras, Carlos Oliva**, Pontificia Universidad Católica de Chile)
- *In vivo* Imaging of Neuronal Activity in the Mouse Somatosensory Cortex Using Multiphoton Microscopy (**Victor Hugo Cornejo**, Pontificia Universidad Católica de Chile)
- Low-Cost Laser Illumination for Microscopy (**Vicente Parot**, Pontificia Universidad Católica de Chile)
- Eye Tracking (Pedro Maldonado, Universidad de Chile)
- Single-Cell Transcriptomics in Neuroscience (**Vinicius Maracaja, Sergio Hidalgo, Patricio Olguín**, Universidad de Chile)